C291 - Spring 25 - Systems Programming with C &
Unix Assignment 2

Q.1 Even or Odd (20 marks)

Write a C program to check whether an integer number is even or odd.

Example 1:
Input a number: 3
Output: 3 is an odd number

Example 2:
Input a number: 12
Output: 12 is an even number

Example 3:
Input a number: 0
Output: 0 is an even number

Note: The solution for this problem should be named as solution_1.c and it should be under C291-Spring-25/assignment2 where assignment2 would be the new directory you would create.

Q.2 Triangle Properties (30 marks)

Write a C program that takes the angles of a triangle as input and checks its properties.
The properties you need to check include:
Validity: For a triangle to be valid the sum of its three angles must be 180 degrees. Besides, every angle must be greater than 0 degrees.
Acute, right or obtuse: If the triangle is valid, then check whether it's an acute, right or obtuse triangle. For a triangle to be acute, all of its angles must be less than 90 degrees; For a triangle to be right-angled, one angle must be equal to 90 degrees; For a triangle to be obtuse, one angle must be greater than 90 degrees.
Isosceles: If the triangle is valid, then check whether it's an isosceles triangle. For a triangle to be isosceles, at least two angles must be the same so two of its sides are also the same.
Equilateral: If the triangle is isosceles, then check whether it's an equilateral triangle. For a triangle to be equilateral, all angles must be the same, and each angle should be exactly 60 degrees. (Only check if the triangle is equilateral when it's isosceles)

Example 1:
Input first angle: 60
Input second angle: 30
Input third angle: 90
Output:
The triangle is valid.
It's a right triangle.
It's not isosceles.

Example 2:
Input first angle: 80
Input second angle: -20
Input third angle: 120
Output:
The triangle is invalid

Note: The solution for this problem should be named as solution_2.c and it should be under C291-Spring-25/assignment2 where assignment2 would be the new directory you would create.

Example 3:
Input first angle: 60
Input second angle: 60
Input third angle: 60
Output:
The triangle is valid.
It's an acute triangle.
It's isosceles.
It's equilateral.

Q.3 Simple Calculator (30 marks)

Write a C program that takes three variables as input from the **scanf()** function to perform a calculation. The three variables are two **double** types for operands and a **char** type for the operator.
The operators you need to implement are:

- + (Addition)
- - (Subtraction)
- * (Multiplication)
- / (Division)
- % (Modulo)
- ^ (Exponentiation)
- r (Root)

For the root operator, $\sqrt{2}$ (square root of 2) will be "2 r 2"; $\sqrt[3]{3}$ (cube root of 9) will be "3 r 9".
In order to use the modulo operator with double types, you need to convert the variables into integers.
When arithmetic errors like dividing by zero occur, you need to print the error instead.
You need to use the **switch** statement for this problem and you are allowed to use functions from the **<math.h>** library.
The result needs to only have **two decimal places**. You can achieve this by using functions from the **<math.h>** library or using **printf()** (The result might be different depending on which method you used, which is fine).

Example 1:
2 ^ 32
Output: 4294967296.00

Example 2:
1 / 0
Output: Error: dividing by zero

Example 3:
3 r 9
Output: 2.08

Note: The solution for this problem should be named as solution_3.c and it should be under C291-Spring-25/assignment2 where assignment2 would be the new directory you would create

Q.4 Date Validity (20 marks)

Write a C program to check if a date is valid or not based on its year, month, and day.

If the year is a **leap year**, February 29 should be valid, otherwise, it's invalid.
Note:
if (year is not divisible by 4) then (it is a common year)
else if (year is not divisible by 100) then (it is a leap year)
else if (year is not divisible by 400) then (it is a common year)
else (it is a leap year)

Example 1:
Enter a year: 2020
Enter a month: 2
Enter a day: 29
Output: The date is valid.

Example 2:
Enter a year: 2021
Enter a month: 2
Enter a day: 29
Output: The date is invalid.

Example 3:
Enter a year: 3030
Enter a month: 15
Enter a day: 50
Output: The date is invalid.

Note: The solution for this problem should be named as solution_4.c and it should be under C291-Spring-25/assignment2 where assignment2 would be the new directory you would create.